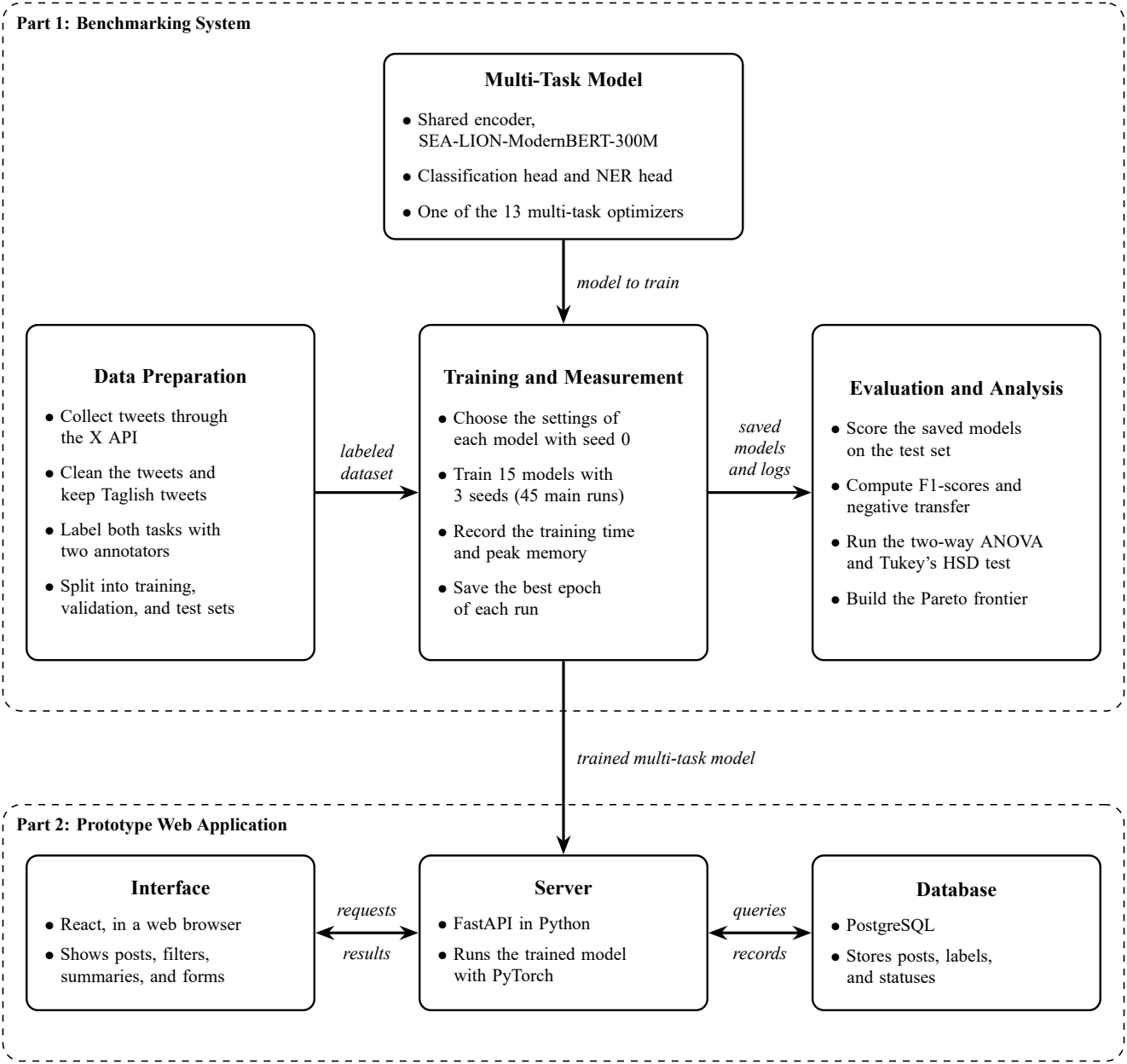

Part 1: Benchmarking System
Multi-Task Model
• Shared encoder, SEA-LION-ModernBERT-300M
• Classification head and NER head
• One of the 13 multi-task optimizers
model to train
Data Preparation
• Collect tweets through the X API
• Clean the tweets and keep Taglish tweets
• Label both tasks with two annotators
• Split into training, validation, and test sets
labeled dataset
Training and Measurement
• Choose the settings of each model with seed 0
• Train 15 models with 3 seeds (45 main runs)
• Record the training time and peak memory
• Save the best epoch of each run
saved models and logs
Evaluation and Analysis
• Score the saved models on the test set
• Compute F1-scores and negative transfer
• Run the two-way ANOVA and Tukey's HSD test
• Build the Pareto frontier
trained multi-task model
Part 2: Prototype Web Application
Interface
• React, in a web browser
• Shows posts, filters, summaries, and forms
requests
results
Server
• FastAPI in Python
• Runs the trained model with PyTorch
queries
records
Database
• PostgreSQL
• Stores posts, labels, and statuses